

For our business, it is therefore sufficient to only compute (minimal) linear pencils for

$$AS\bar{R}S^\top\bar{M}_1, \quad (38)$$

$$A\bar{M}_1S\bar{R}S^\top BS\bar{R}S^\top, \quad (39)$$

$$A\bar{M}_1S\bar{R}S^\top BS\bar{R}S^\top\bar{M}_1, \quad (40)$$

$$A\bar{M}_1S\bar{R}S^\top BS\bar{R}S^\top\bar{M}_2, \quad (41)$$

$$\text{where } \bar{M}_k := \Sigma_k^{1/2} Z_k^\top Z_k \Sigma_k^{1/2}, \bar{R} := (S^\top \bar{M} S + I_m)^{-1}, \bar{M} := \bar{M}_1 + \bar{M}_2.$$

Observe that without the S matrix (i.e taking $m = d$ and $S = I_d$), the four matrix expressions above reduce to what we had in the classical case.

D Deterministic Equivalents

D.1 Classical Linear Model

Note that the weights $\hat{w}$ of the model $\hat{f}_{CL}$ given in (3) can be written explicitly as $\hat{w} = RX^\top Y$, where $R := (X^\top X + n\lambda I_d)^{-1} = (X_1^\top X_1 + X_2^\top X_2 + n\lambda I_d)^{-1}$, a random matrix. Its test error $E_{test}(\hat{f}_{CL})$ writes $E_{test}(\hat{f}_{CL}) = \mathbb{E}_{X,Y}[(\hat{f}_{CL}(x) - x^\top w_1^*)^2] = \mathbb{E}_{X,Y}[\|\hat{w} - w_1^*\|_{\Sigma_1}^2]$. In Proposition 1, we shall show that the RHS in the above can be decomposed into a sum of simply random quantities of the form $r_j^{(k)}(A)$ and $r_j^{(k)}(A, B)$ that we now describe and analyze.

Let A and B be $d \times d$ positive-definite matrices with well-behaved spectral (this will be made precise latter) and let $\lambda > 0$. In analyzing the bias-variance decomposition of the test error, we are ultimately led to consider the following quantities

$$r_j^{(1)}(A) := \mathbb{E} \operatorname{tr} AM_j(M + \lambda I_d)^{-1}, \quad (42)$$

$$r_j^{(2)}(A, B) := \mathbb{E} \operatorname{tr} A(M + \lambda I_d)^{-1} B(M + \lambda I_d)^{-1}, \quad (43)$$

$$r_j^{(3)}(A, B) := \mathbb{E} \operatorname{tr} AM_j(M + \lambda I_d)^{-1} B(M + \lambda I_d)^{-1} M_j, \quad (44)$$

$$r_j^{(4)}(A, B) := \mathbb{E} \operatorname{tr} AM_j(M + \lambda I_d)^{-1} B(M + \lambda I_d)^{-1}, \quad (45)$$

where we recall that $M := M_1 + M_2$ and $M_j := X_j^\top X_j/n$.

Let (e_1, e_2) be the unique negative solution to the following pair of fixed-point equations

$$e_1 = \frac{1}{1 + \phi \operatorname{tr} \Sigma_1 K^{-1}}, \quad e_2 = \frac{1}{1 + \phi \operatorname{tr} \Sigma_2 K^{-1}}, \quad \text{with } K := p_1 e_1 \Sigma_1 + p_2 e_2 \Sigma_2 + \lambda I_d. \quad (46)$$

Also, define (u_1, u_2) to be the unique positive solution to the pair of fixed-point equations

$$u_1 = \phi e_1^2 \bar{\operatorname{tr}} \Sigma_1 L' K^{-2}, \quad u_2 = \phi e_2^2 \bar{\operatorname{tr}} \Sigma_2 L' K^{-2}, \quad \text{with } L' := p_1 u_1 \Sigma_1 + p_2 u_2 \Sigma_2 + \lambda B. \quad (47)$$

Consider the following deterministic matrices

$$\begin{aligned} C_j &:= p_j e_j^2 (B + p_{j'} u_{j'} \Sigma_{j'}) \Sigma_1 + u_1 (p_{j'} e_{j'} \Sigma_{j'} + \lambda I_d)^2, \\ D_j &:= e_j B - \lambda u_j I_d + p_{j'} (e_j u_{j'} - e_{j'} u_j) \Sigma_{j'}, \end{aligned} \quad (48)$$

where $1' := 2$ and $2' = 1$.

The following will be crucial for proving Theorem 1 and its corollaries.

Proposition 2. *In the proportionate scaling limit (4), it holds for $j = 1, 2$ that*

$$r_j^{(1)}(A) \simeq p_j e_j \operatorname{tr} A \Sigma_j K^{-1}, \quad (49)$$

$$r_j^{(2)}(A, B) \simeq \operatorname{tr} A L' K^{-2}, \quad (50)$$

$$r_j^{(3)}(A, B) \simeq p_j \operatorname{tr} A \Sigma_j C_j K^{-2}, \quad (51)$$

$$r_j^{(4)}(A, B) \simeq p_j \operatorname{tr} A \Sigma_j D_j K^{-2}. \quad (52)$$

D.2 Random Projections

For $d \times d$ deterministic matrices A and B , define the following quenched quantities

$$\begin{aligned} r_j^{(1)}(A) &:= \mathbb{E} \operatorname{tr} ASRS^\top M_j, \quad r_j^{(3)}(A, B) := \mathbb{E} \operatorname{tr} AM_jSR^\top SBSRS^\top M_j, \\ r_j^{(4)}(A, B) &:= \mathbb{E} \operatorname{tr} AM_jSR^\top SBSRS^\top, \quad r_j^{(5)}(A, B) := \mathbb{E} \operatorname{tr} AM_1SR^\top SBSRS^\top M_2, \end{aligned} \quad (53)$$

where we recall that $R := (S^\top MS + \lambda I_m)^{-1}$, $M := M_1 + M_2$, $M_j := X_j^\top X_j/n$. These quantities will be useful because we may write

$$\begin{aligned} V_k &= \sum_j \sigma_j^2 \frac{1}{n} \mathbb{E} \operatorname{tr} M_j SRS^\top \Sigma_k SRS^\top = \sum_{j=1}^2 \frac{\sigma_j^2}{n} r_j^{(4)}(I_d, \Sigma_k), \\ B_k &= \operatorname{tr} \Gamma \Sigma_k + \mathbb{E} \operatorname{tr} \Gamma M SRS^\top \Sigma SRS^\top M - 2 \operatorname{tr} \Gamma \Sigma_k SRS^\top M + \operatorname{tr} \Delta M_2 SRS^\top \Sigma_k SRS^\top M_2 \\ &= \operatorname{tr} \Gamma \Sigma_k + 2r^{(5)}(\Gamma, \Sigma_k) + r_1^{(3)}(\Gamma, \Sigma_k) + r_2^{(3)}(\Gamma, \Sigma_k) - 2r_1^{(1)}(\Gamma \Sigma_k) - 2r_2^{(1)}(\Gamma \Sigma_k) + r_2^{(3)}(\Delta, \Sigma_k). \end{aligned}$$

Each term in the above decomposition can now be analyzed via operator-valued free-probability theory. The following proposition will be heavily exploited in the prove of Theorem 2.

Proposition 3. *In the proportionate scaling limit (6), it holds that*

$$\begin{aligned} r_j^{(1)}(A) &\simeq p_j \gamma \tau e_j \operatorname{tr} A \Sigma K^{-1}, \quad r_j^{(3)}(A, \Sigma) \simeq p_j \operatorname{tr} A \Sigma C_j K^{-2}, \\ r_j^{(4)}(A, \Sigma) &\simeq p_j \gamma \operatorname{tr} A \Sigma D K^{-2}, \quad r_j^{(5)}(A, \Sigma) \simeq p_1 p_2 \gamma \operatorname{tr} A \Sigma^2 E K^{-2}, \end{aligned} \quad (54)$$

where the constants e_1 and e_2 and the matrices C_1 , C_2 , D , and E are as in Theorem 2.

D.3 Proof of Proposition 2

WLOG, we only consider the case $j = 1$, and suppress this subscript henceforth from all the $r_j^{(k)}$'s.

Computing $r_j^{(1)}$. We only do $j = 1$ as $j = 2$ is completely analogous. Consider the following 9×9 block matrix

$$Q = \begin{bmatrix} I_d & 0 & -\Sigma_1^{1/2} & 0 & 0 & 0 & 0 & 0 & 0 \\ -A & I_d & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I_d & -\frac{Z_1^\top}{\sqrt{n\lambda}} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I_{n_1} & -\frac{Z_1}{\sqrt{n\lambda}} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\Sigma_1^{1/2} & 0 & 0 & 0 \\ 0 & 0 & \Sigma_1^{1/2} & 0 & 0 & I_d & \Sigma_2^{1/2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & I_d & -\frac{Z_2^\top}{\sqrt{n\lambda}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & I_{n_2} & -\frac{Z_2}{\sqrt{n\lambda}} \\ 0 & 0 & 0 & 0 & 0 & -\Sigma_2^{1/2} & 0 & 0 & I_d \end{bmatrix}. \quad (55)$$

This is a (minimal) linear pencil for the random matrix $R = AM_1(M + \lambda I_d)^{-1}$; indeed it is easy to verify that $R = Q^{-1}[1, 5]/\lambda$ (using zero-based indexing). It follows that in the asymptotic limit, one has

$$r^{(1)}/d = \mathbb{E} \bar{\operatorname{tr}} R \simeq G_{1,5}, \quad (56)$$

where $G = (\operatorname{id} \otimes \mathbb{E} \bar{\operatorname{tr}})[Q^{-1}] \in M_9(\mathbb{C})$ is the matrix containing the limiting expected values of the normalized traces of the blocks of each of the $9 \times 9 = 81$ blocks of Q^{-1} (we define the trace of each rectangular block as zero). Using classical operator-valued free probability theory (OVFPT) Mingo

& Speicher (2017), we have the following fixed-point equations which define $G_{1,5}$ implicitly

$$G_{1,5} = p_1 G_{3,3} \bar{\text{tr}} A \Sigma_1 (\lambda I_d + p_1 G_{3,3} \Sigma_1 + p_2 G_{7,7} \Sigma_2)^{-1}, \quad (57)$$

$$G_{3,3} = \frac{\lambda}{\lambda - \phi G_{4,2}}, \quad (58)$$

$$G_{7,7} = \frac{\lambda}{\lambda - \phi G_{8,6}}, \quad (59)$$

$$G_{4,2} = -\lambda \bar{\text{tr}} \Sigma_1 (\lambda I_d + p_1 G_{3,3} \Sigma_1 + p_2 G_{7,7} \Sigma_2)^{-1}, \quad (60)$$

$$G_{8,6} = -\lambda \bar{\text{tr}} \Sigma_2 (\lambda I_d + p_1 G_{3,3} \Sigma_1 + p_2 G_{7,7} \Sigma_2)^{-1}. \quad (61)$$

We deduce that $G_{3,3} = e_1$, $G_{7,7} = e_2$, and

$$r^{(1)}/d = G_{1,5} = p_1 e_1 \bar{\text{tr}} A \Sigma_1 (\lambda I_d + p_1 e_1 \Sigma_1 + p_2 e_2 \Sigma_2)^{-1},$$

where (e_1, e_2) is the unique pair of nonnegative solutions to the system of equations

$$e_1 = \frac{1}{1 + \phi \bar{\text{tr}} \Sigma_1 (\lambda I_d + p_1 e_1 \Sigma_1 + p_2 e_2 \Sigma_2)^{-1}}, \quad (62)$$

$$e_2 = \frac{1}{1 + \phi \bar{\text{tr}} \Sigma_2 (\lambda I_d + p_1 e_1 \Sigma_1 + p_2 e_2 \Sigma_2)^{-1}}. \quad (63)$$

Putting things together gives

$$r^{(1)} \simeq d \cdot G_{1,5} = p_1 e_1 \text{tr} A \Sigma_1 (p_1 e_1 \Sigma_1 + p_2 e_2 \Sigma_2 + \lambda I_d)^{-1} = p_1 \text{tr} A \Sigma_1 K^{-1}.$$

In particular, in the limit $p_2 \rightarrow 0^+$ (i.e single data source), the first equation becomes

$$\begin{aligned} 1 - \lambda/\kappa_1 &= 1 - \eta_1 \lambda = \phi_1 \eta_1 \bar{\text{tr}} \Sigma_1 (I_d + p_1 \eta_1 \Sigma_1)^{-1} \\ &= \phi_1 \bar{\text{tr}} \Sigma_1 (\kappa_1 I_d + \Sigma_1)^{-1}, \end{aligned}$$

or equivalently,

$$\kappa_1 - \lambda \simeq \kappa_1 \frac{\text{df}_1(\kappa_1; \Sigma_1)}{n_1}. \quad (64)$$

Furthermore, $r^{(1)}$ is now given by

$$r^{(1)} \simeq e_1 \text{tr} A \Sigma_1 (e_1 \Sigma_1 + \lambda I_d)^{-1} = \text{tr} A \Sigma_1 (\Sigma_1 + \kappa_1 I_d)^{-1}. \quad (65)$$

Computing $r^{(4)}$. Here, the minimal linear pencil for the random matrix involved $R = AM_1(M + \lambda I_d)^{-1}B(M + \lambda I_d)^{-1}$ is a 16×16 block matrix Q (not shown here¹) such that $R = Q^{-1}[1, 9]/\lambda$. Thus, $r^{(4)}/d \simeq G_{1,16}/\lambda$, where $G = (\text{id} \otimes \mathbb{E} \bar{\text{tr}})[Q^{-1}] \in M_{16}(\mathbb{C})$.

First consider the special case $p_2 \rightarrow 0^+$ (i.e n_2 is negligible compared to n_1). The fixed-point equations defining $G_{1,9}$ are given by

$$G_{1,9} = \lambda \bar{\text{tr}} A \Sigma_1 (G_{3,3} B + G_{3,11} I_d) (\lambda I_d + G_{3,3} \Sigma_1)^{-1} (\lambda I_d + G_{11,11} \Sigma_1)^{-1}, \quad (66)$$

$$G_{3,3} = \frac{\lambda}{\lambda - \phi G_{4,2}}, \quad (67)$$

$$G_{11,11} = \frac{\lambda}{\lambda - \phi G_{12,10}}, \quad (68)$$

$$G_{3,11} = \frac{\lambda \phi G_{4,10}}{(\lambda - \phi G_{4,2})(\lambda - \phi G_{12,10})} = \frac{\phi G_{3,3} G_{11,11} G_{4,10}}{\lambda}, \quad (69)$$

$$G_{12,10} = -\lambda \bar{\text{tr}} \Sigma_1 (\lambda I_d + G_{11,11} \Sigma_1)^{-1}, \quad (70)$$

$$G_{4,10} = -\lambda \bar{\text{tr}} \Sigma_1 (\lambda B - G_{3,11} \Sigma_1) (\lambda I_d + G_{3,3} \Sigma_1)^{-1} (\lambda I_d + G_{11,11} \Sigma_1)^{-1}, \quad (71)$$

$$G_{4,2} = -\lambda \bar{\text{tr}} \Sigma_1 (\lambda I_d + G_{3,3} \Sigma_1)^{-1}. \quad (72)$$

¹All the linear pencils in this work are very big and are omitted for brevity.